

Week 2 – Programming Basics

1. Swap two variables without using a temporary variable

Description: Write a program to swap the values of two variables (e.g., a and b) without using a third variable.

Example:

Input:

a = 5, b = 10

Output:

a = 10, b = 5

2. Find the largest and smallest of three numbers

Description: Read three integers and print the largest and smallest.

Example:

Input:

3, 9, 5

Output:

Largest: 9, Smallest: 3

3. Calculate the area and perimeter of a rectangle

Description: Given length and width, calculate and display both area and perimeter.

Example:

Input:

length = 4, width = 6

Output:

Area: 24, Perimeter: 20

4. Temperature converter (Celsius ↔ Fahrenheit)

Description: Convert temperature from Celsius to Fahrenheit and vice versa.

Formula:

$$F = (C \times 9/5) + 32$$

$$C = (F - 32) \times 5/9$$

Example:

Input:

C = 25

Output:

F = 77

5. Compute Simple and Compound Interest

Description: Given Principal (P), Rate (R), and Time (T), compute:

Simple Interest = $(P \times R \times T) / 100$

Compound Interest = $P \times ((1 + R/100) ^ T - 1)$

Example:

Input:

P = 1000, R = 5, T = 2

Output:

SI = 100, CI $\approx$ 102.5

6. Convert total seconds into hours, minutes, and seconds

Description: Convert a given number of seconds into hours, minutes, and seconds format.

Example:

Input:

3665

Output:

1 hr, 1 min, 5 sec

7. Check whether a number is even or odd

Description: Determine if a number is divisible by 2.

Example:

Input:

7

Output:

Odd

8. Determine if a number is positive, negative, or zero

Description: Use if-else statements to categorize a number.

Example:

Input:

-3

Output:
Negative

9. Grade Calculator (0–100 → A–F)

Description: Input a student's marks (0–100) and print the grade based on a scale.

Example:

Input:

85

Output:

B

10. Leap Year Checker

Description: Determine if a given year is a leap year.

Rule: Year divisible by 4 → leap year, except centuries not divisible by 400.

Example:

Input:

2000

Output:

Leap year

Input:

1900

Output:

Not leap year

11. Simple Calculator using switch-case

Description: Input two numbers and an operator (+, -, *, /) and compute the result.

Example:

Input:

5, 3, +

Output:

8

12. Traffic Light System using switch-case

Description: Input a color (Red, Yellow, Green) and display an action (Stop, Ready, Go).

Example:

Input:

Green

Output:

Go

13. Month name & number of days using switch-case

Description: Input a month number (1–12) and display its name and number of days.

Example:

Input:

2

Output:

February, 28 or 29 days

14. Check character type

Description: Input a character and determine if it's a vowel, consonant, digit, or special symbol.

Example:

Input:

a

Output:

Vowel

15. Triangle validity and type

Description: Given three sides, determine if they form a valid triangle and if it's equilateral, isosceles, or scalene.

Rule: Sum of any two sides > third side.

Example:

Input:

3, 4, 5

Output:

Valid triangle, Scalene

16. Print numbers from 1 to 100

Description: Use a loop to print numbers sequentially.

Example:

Output:

1, 2, 3, ..., 100

17. Sum of first N natural numbers

Description: Calculate and print the sum using a loop.

Example:

Input:

5

Output:

15

18. Multiplication table of a number

Description: Print the multiplication table for a given number (up to 10 or N).

Example:

Input:

5

Output:

$5 \times 1 = 5$

$5 \times 2 = 10$

...

$5 \times 10 = 50$

19. Reverse a number

Description: Reverse the digits of an integer.

Example:

Input:

1234

Output:

4321

20. Count digits in a number

Description: Find how many digits are in a number.

Example:

Input:

786

Output:

3

21. Sum of digits

Description: Calculate the sum of all digits of a number.

Example:

Input:

123

Output:

6

22. Print even numbers in a range

Description: Input start and end values, print all even numbers between them.

Example:

Input:

1, 10

Output:

2, 4, 6, 8, 10

23. Fibonacci series up to N terms

Description: Generate the Fibonacci sequence (0, 1, 1, 2, 3, 5, 8...) up to N terms.

Example:

Input:

7

Output:

0, 1, 1, 2, 3, 5, 8

24. Find GCD of two numbers

Description: Compute the greatest common divisor using repeated subtraction or the Euclidean algorithm.

Example:

Input:

12, 18

Output:

6

25. Find LCM of two numbers

Description: Compute least common multiple using the relationship: $\text{LCM}(a, b) = (a \times b) / \text{GCD}(a, b)$

Example:

Input:

12, 18

Output:

36